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DRIVER ASSISTANCE SYSTEM

Abstract

The driver assistance system acquires the vehicle control content in a case where it is assumed that the automated driving control is performed during the manual driving of the vehicle by the driver. The driver assistance system acquires the driver operation content, which is the content of the driving operation of the vehicle by the driver, during the manual driving of the vehicle by the driver. The driver assistance system notifies the driver of feedback related to the driving operation on the basis of a result of the comparison between the vehicle control content and the driver operation content.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-031091 filed on Mar. 1, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to technology for notifying a driver of a vehicle with feedback related to driving operations.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2020-140159 (JP 2020-140159 A) discloses an instruction system for instructing a driver that is driving a vehicle regarding driving operations. The instruction system notifies the driver, who is driving, of operation contents determined based on a current state of the vehicle and a past traveling record thereof.

SUMMARY

[0004] Feedback (notification) based on the traveling record is not based on an actual state of the object vehicle, and accordingly may not necessarily be optimal feedback. In particular, it is expected that validity of the feedback will be low when there are few travel records accumulated regarding that traveling road.

[0005] One object of the present disclosure is to perform notification of appropriate feedback, with respect to driving operation contents of the driver, by a method that is not dependent on the travel record.

[0006] A first aspect relates to a driver assistance system for a driver of a vehicle. The driver assistance system includes a control device that is equipped with a function of automated driving control.

[0007] The control device acquires a vehicle control content when assumption is made that the automated driving control will be performed, during manual driving of the vehicle by the driver.

[0008] The control device acquires a driver operation content that is a content of a driving operation of the vehicle by the driver, during manual driving of the vehicle by the driver.

[0009] The control device notifies the driver of feedback regarding the driving operation, based on a result of comparison between the vehicle control content and the driver operation content.

[0010] According to the first aspect, the driver assistance system notifies the driver of the feedback based on the comparison between the vehicle control content and the operation content by the driver, when it is assumed that automated driving control will be performed, during manual driving of the vehicle by the driver. The vehicle control content is determined based on the current state of the vehicle rather than the traveling record, and accordingly the driver can be notified of appropriate feedback without depending on the traveling record.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0012] FIG. 1 is a conceptual diagram for explaining an outline of a driver assistance system according to the present embodiment;

[0013] FIG. 2 is a block diagram illustrating an example of driving environment information according to the present embodiment;
[0014] FIG. 3 is a schematic diagram illustrating a state in which the driver assistance system according to the present embodiment notifies the driver of feedback;
[0015] FIG. 4 is a more detailed representation of the situation in FIG. 3;
[0016] FIG. 5 is a block-diagram illustrating a detailed configuration of the driver assistance system according to the present embodiment; and
[0017] FIG. 6 is a block diagram illustrating a modification of the driver assistance system according to the present embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0018] Embodiments of the present disclosure will be described with reference to the accompanying drawings.

1. Driver Assistance System

[0019] FIG. 1 is a conceptual diagram for explaining an outline of a driver assistance system **10** according to the present embodiment. The driver assistance system **10** controls the vehicle **1**. Typically, the driver assistance system **10** is mounted on the vehicle **1**.

[0020] The driver assistance system **10** includes a sensor group **20**, an HMI (Human Machine Interface) unit **30**, a traveling device **50**, a communication device **60**, and a control device **70**. At least the sensor group **20**, HMI unit **30**, the traveling device **50**, and the communication device **60** are mounted on the vehicle **1**.

[0021] The traveling device **50** (actuator) includes a steering device, a driving device, and a braking device. The steering device will turn the wheel. For example, the steering device includes a power steering (EPS: Electric Power Steering) device. The driving device is a power source that generates a driving force. Examples of the driving device include an engine, an electric motor, and an in-wheel motor. The braking device generates a braking force.

[0022] The sensor group **20** includes a recognition sensor **21**, a vehicle state sensor **22**, a position sensor **23**, and a driver operation sensor **24**. The recognition sensor **21** recognizes (detects) a situation around the vehicle **1**. Examples of the recognition sensor **21** include a camera LIDAR (Laser Imaging Detection and Ranging), a radar, and the like. The vehicle state sensor **22** detects the state of the vehicle **1**. For example, the vehicle state sensor **22** includes a speed sensor, an acceleration sensor, a yaw rate sensor, a steering angle sensor, and the like. The position sensor **23** detects the position and the azimuth of the vehicle **1**. For example, the position sensor **23** includes a GNSS (Global Navigation Satellite System).

[0023] The driver operation sensor **24** acquires the presence or absence of a driving operation by the driver of the vehicle **1** and the operation amount thereof. For example, the driver operation sensor **24** includes a throttle valve opening degree sensor, an engine speed sensor, a braking force sensor, an EPS torque sensor, and the like. While the driver is performing the manual operation, the presence or absence of the driving operation by the driver and the operation amount thereof can be acquired on the basis of changes in the throttle valve opening degree, the engine speed, the braking force, EPS torque, and the like. As another example, the driver operation sensor **24** may include a steering sensor, an accelerator pedal sensor, and a brake pedal sensor. The steering sensor detects a steering amount of a steering wheel. The accelerator pedal sensor detects a stroke amount of the accelerator pedal. The brake pedal sensor detects a stroke amount of the brake pedal.

[0024] HMI unit **30** is an interface for providing information to a driver of the vehicle **1** and receiving information from the driver. Specifically, HMI unit **30** includes an inputting device and an outputting device. Examples of the input device include a touch panel, a switch, and a microphone. Examples of the output device include a display device, a speaker, an in-vehicle LED, and the like. Examples of the display device include a liquid crystal panel and an organic EL panel. HMI unit **30** may include a HUD (Head Up Display), a vibrator, and the like.

[0025] The communication device **60** communicates with the outside via a communication

network. Examples of the communication scheme include mobile communication such as 5G, and radio LAN.

[0026] The control device **70** is a computer that controls the vehicle **1**. Typically, the control device **70** is mounted on the vehicle **1**. However, a part of the control device **70** may be disposed in an external device, and the vehicle **1** may be controlled remotely. The control device **70** executes various processes. For example, the control device **70** includes a processing circuitry such as a CPU (Central Processing Unit). The processing circuit may also be referred to as a processor. The control device **70** includes one or more storage devices **72** (hereinafter simply referred to as storage devices **72**). The storage device **72** stores various types of information. Examples of the storage device **72** include volatile memory, non-volatile memory, HDD (Hard Disk Drive), SSD (Solid State Drive), and the like.

[0027] The control program **80** is a computer program for controlling the vehicle **1**. The function of the control device **70** may be realized by cooperation of the control device **70** that executes the control program **80** and the storage device **72**. The control program **80** is stored in the storage device **72**. Alternatively, the control program **80** may be recorded in a computer-readable recording medium.

[0028] The control device **70** acquires the driving environment information **90** indicating the driving environment of the vehicle **1**. The driving environment information **90** is stored in the storage device **72**.

[0029] FIG. **2** is a block diagram illustrating an example of the driving environment information **90**. The driving environment information **90** includes map information **91**, surrounding situation information **92**, vehicle state information **93**, and vehicle position information **94**.

[0030] The map information **91** includes a general navigation map. The map information **91** may indicate a lane arrangement or a road shape. The map information **91** may include positional information such as a structure, a traffic light, a sign, and the like. The control device **70** acquires map information **91** of a necessary area from the map database. The map database may be stored in the storage device **72** or may be stored in a map management device outside the vehicle **1**. In the latter case, the control device **70** communicates with the map management device via the communication device **60**, and acquires necessary map information **91**.

[0031] The surrounding situation information **92** is information obtained based on a recognition result by the recognition sensor **21**, and indicates a situation around the vehicle **1**. The control device **70** recognizes a situation around the vehicle **1** using the recognition sensor **21**, and acquires the surrounding situation information **92**. For example, the surrounding situation information **92** includes image IMG captured by the camera. Alternatively, the surrounding situation information **92** includes point cloud information obtained by LIDAR.

[0032] The surrounding situation information **92** further includes object information OBJ related to objects (targets) around the vehicles **1**. Examples of the object include pedestrians, bicycles, motorcycles, other vehicles (preceding vehicles, parked vehicles, and the like), white lines, traffic lights, structures (e.g., poles, pedestrian bridges), signs, obstacles, and the like. The object information OBJ indicates the relative position and the relative velocity of the object with respect to the vehicle **1**. For example, an object can be identified and the relative position of the object can be calculated by analyzing the image IMG obtained by the camera. It is also possible to identify an object based on the point cloud data obtained by LIDAR, and to acquire the relative position and the relative velocity of the object. The control device **70** may track the recognized object. The object information OBJ then also includes the trajectory information of the recognized object.

[0033] The vehicle state information **93** is information detected by the vehicle state sensor **22** and indicates the state of the vehicle **1**. The state of the vehicle **1** includes vehicle speed, acceleration, yaw rate, steering angle, and the like. The control device **70** acquires the vehicle state information **93** from the vehicle state sensor **22**. The vehicle state information **93** may indicate a driving state (automatic driving/manual driving) of the vehicle **1**.

[0034] The vehicle position information **94** is information indicating the current position of the vehicle **1**. The control device **70** acquires the vehicle position information **94** from the detection result by the position sensor **23**. Further, the control device **70** may acquire the highly accurate vehicle-position information **94** by a well-known self-position estimation process (Localization) using the object information OBJ and the map information **91**.

[0035] Further, the control device **70** executes vehicle travel control for controlling the travel of the vehicle **1**. The vehicle travel control includes steering control, acceleration control, and deceleration control. The control device **70** executes vehicle travel control by controlling the traveling device **50** (a steering device, a drive device, and a braking device). More specifically, the control device **70** calculates a control amount (actuator control amount) of the traveling device **50**, and controls the traveling device **50** in accordance with the actuator control amount.

[0036] Further, the control device **70** performs automated driving control for controlling autonomous driving of the vehicle **1**. Here, the autonomous driving means that at least a part of the steering, the acceleration, and the deceleration of the vehicle **1** is automatically performed independently of the operation of the driver. As an example, level 3 or more automated driving may be performed. The control device **70** generates a travel plan of the vehicle **1** based on the driving environment information **90**. Examples of the traveling plan include maintaining the current traveling lane, changing lanes, turning right and left, avoiding collision with an object, and the like. More specifically, the travel plan includes a route plan and a speed plan of the vehicle **1**. The route plan is a set of target positions of the vehicle **1**. The velocity plan is a set of target velocities for each target position. The combination of path planning and velocity planning is also referred to as "target trajectory". That is, the target trajectory includes a target position and a target speed of the vehicle **1**. The control device **70** performs vehicle travel control so that the vehicle **1** follows the target trajectory TR. The control device **70** may perform automated driving control while performing optimization by model-predictive control (MPC: Model Predictive Control).

2. Feedback Notification

2-1. Overview of Feedback Notification

[0037] FIG. **3** is a schematic diagram illustrating a state in which the driver assistance system **10** notifies the driver of the feedback FB. It is assumed that the vehicle **1** includes the driver assistance system **10** and is in a manual driving state by a driver. When the vehicle **1** enters the curve from the straight section, in order to safely exit the curve, the driver needs to decelerate the vehicle **1** by operating the brake. The content of the optimum braking operation in this case depends on the current state of the vehicle **1** (vehicle speed, acceleration, yaw rate, steering angle, etc.). The braking operation content includes at least a timing of the braking operation and a braking operation amount. The driver performs a braking operation that he/she thinks is appropriate, but it is not known how much the braking operation differs from the optimum operation.

[0038] Therefore, the driver assistance system **10** according to the present disclosure acquires the optimum driving operation content by calculation based on the automated driving control. The driver assistance system **10** compares the acquired optimum driving operation content with the actual driving operation content of the driver, and notifies a feedback FB of the driving operation by the driver. Since the feedback FB is based on the present condition of the vehicle **1**, it can be said that the feedback is more valid than the feedback based on the previous travel records. The driving operation to be the target of the feedback FB of the driver assistance system **10** includes an accelerator operation, a steering operation, and the like in addition to the braking operation exemplified above. Specific embodiments thereof will be described below.

[0039] The control device **70** acquires the vehicle control content when it is assumed that the automated driving control is performed. In FIG. **3**, the control device **70** included in the driver assistance system **10** generates the target trajectory TR. The control device **70** performs automated driving control so that the vehicles **1** follow the target trajectory TR. As described above, since the target trajectory is generated based on the driving environment information **90**, it can be said that

the target trajectory is an optimal trajectory that is generated based on the current state of the vehicle **1**.

[0040] The control device **70** acquires the vehicle control content (hereinafter, referred to as “assumed control content”) in a case where it is assumed that the automated driving control is performed. In the embodiment of FIG. **3**, the vehicle control content required when the vehicle **1** is assumed to travel following the target trajectory TR by automated driving control is acquired. The vehicle control content includes a timing for controlling the traveling device **50** (actuator) and an amount for controlling.

[0041] The control device **70** calculates the operation content of the traveling device **50** (actuator) corresponding to the assumed control content. The operation content of the actuator corresponding to the assumed control content can be referred to as the optimum operation content when the driver performs the driving operation. Therefore, in the following description, the operation content of the actuator corresponding to the assumed control content is referred to as “optimum operation content”. In the automated driving control, the control device **70** controls the behavior of the vehicle **1** by controlling the actuator so that the vehicle **1** follows the target trajectory TR. Therefore, it can be said that the optimum operation content is included in the assumed control content.

[0042] The control device **70** acquires, via the driver operation sensor **24**, the driver operation content that is the content of the driving operation of the vehicle **1** by the driver. The driver operation content is an operation content of the traveling device **50** (actuator) actually operated by the driver, and includes a timing at which a driving operation is performed by the driver and an amount of the operation.

[0043] The control device **70** compares the optimum operation content with the driver operation content. The control device **70** is configured to notify the driver of the feedback FB related to the driving operation. Examples of the content of the feedback FB include an operation timing of the driver being too early/too late, or an operation amount of the driver being too small/too large. The feedback FB is notified via HMI unit **30**, for example. Audio notifications and visual displays (meters, displays, in-vehicle LED, and the like) using speakers correspond to notifications via HMI unit **30**. Alternatively, the notification may be made by vibrating the steering wheel or the seat in a specific pattern.

2-2. Effect

[0044] As described above, the driver assistance system **10** notifies the driver of the feedback FB related to the driving operation of the driver based on the present condition of the vehicle **1**. JP 2020-140159 A discloses a system in which feedback is provided with reference to past travel records of the same travel path. However, the travel record referred to in such a system is not necessarily an optimal feedback because it is not based on the actual state of the target vehicle. In particular, it is expected that the validity of the feedback will be further reduced if there are few travel records accumulated for the travel path. On the other hand, the driver assistance system **10** notifies the feedback FB based on the actual vehicle condition of the vehicle **1**. Accordingly, an appropriate feedback FB can be notified without depending on the travel recording.

[0045] The driver assistance system **10** is particularly effective for guidance of a driving trainer. When the driving operation of the driving trainer does not conform to the optimum operation content, the driver assistance system **10** notifies the driving trainer of feedback FB regarding whether the operation timing is good or bad and the strength of the operation. The driver can train while confirming the difference between the operation of the driver and the optimum operation content, so that it is possible to expect improvement of the efficient driving technology. The location to which the driver assistance system **10** can be applied is not limited to a specific area such as a driving training area.

3. Viewpoint of Manipulated Variable and Manipulation Timing

[0046] FIG. **4** is a diagram illustrating the situation in FIG. **3** in more detail. In FIG. **4**, the vehicle

control timing (brake control timing) based on the assumed control content is time **t1**. Further, the vehicle control amount (brake control amount) based on the assumed control content at the time **t1** is the control amount **C1**. That is, assuming that the vehicle **1** is under automated driving control, the control device **70** performs braking control of the control amount **C1** at the time **t1** in order to cause the vehicle **1** to follow the target trajectory **TR**.

[0047] The control device **70** calculates the optimum operation content that is the operation content of the actuator corresponding to the assumed control content. In FIG. **4**, the optimum actuator operating amount **AC1** is calculated based on the control amount **C1**. That is, the control device **70** calculates the optimum manipulation content (**t1**, **AC1**) from the hypothetical control content (**t1**, **C1**). Therefore, it can be said that the closer the driver operation content by the driver is to the optimum operation content calculated by the control device **70**, the closer the driving operation of the driver is to the ideal.

[0048] The driver assistance system **10** compares the driver operation content (including the driving operation timing and the driving operation amount) with a predetermined range including the optimum operation content, thereby branching the presence or absence of the feedback **FB** and the content thereof. When the driver operation content is out of the predetermined range, a feedback **FB** indicating that the driver operation is not appropriate is notified. When the driver operation content is within a predetermined range, a feedback **FB** indicating that the driver operation is appropriate may be notified.

[0049] In FIG. **4**, the optimum operation **OPr** indicates an optimum operation content (**t1**, **AC1**). The temporal range including the time **t1** is defined as the first range **R1**. The range of the operating amount including the actuator operating amount **AC1** is set as the second range **R2**. When the driver performs the operation **OP1**, the operation timing is within the first range **R1** (appropriate), and the operation amount is located outside the second range **R2** (excessive). A feedback **FB** indicating that the manipulated variable is excessive is notified. At this time, the operation timing may be fed back together to the effect that it is appropriate. When the driver performs the operation **OP2**, the operation timing is outside the first range **R1** (slow), and the operation amount is outside the second range **R2** (too small). A feedback **FB** indicating that the operation timing is slow and the operation amount is too small is notified. When the driver performs the operation **OP3**, the operation timing is within the first range and the operation amount is within the second range. In such cases, a feedback **FB** (e.g., **GOOD** etc.) indicating that the driver-operated content is appropriate may be notified. In addition, when the operation of the driver is not detected from the time **t1** for a predetermined period of time, the control device **70** may consider that there is no operation by the driver and notify the driver of a feedback **FB** that prompts the driver to perform the operation.

[0050] Note that the first range and the second range do not need to be centered on the optimum operation content. For example, in the braking operation, if the operation timing is early, it means that the vehicle **1** starts to decelerate quickly. This can be said to be the shift of the “safety side”. On the other hand, when the operation timing of the brake is slow, the vehicle **1** starts to decelerate slowly, so that it can be said that the shift is on the “dangerous side”. Therefore, it is reasonable to set the allowable range for the operation on the safety side to be wider than the optimum operation content, and the allowable range for the operation on the dangerous side to be narrower than the optimum operation content.

4. Configuration Example of the Driver Assistance System

[0051] FIG. **5** is a block diagram illustrating a detailed configuration example of the driver assistance system **10**.

[0052] The control content acquisition unit **71** acquires various pieces of information from the recognition sensor **21**, the vehicle state sensor **22**, and the position sensor **23**. The target trajectory generation unit **71a** generates a target trajectory **TR** based on the data acquired by the control content acquisition unit **71**. The control content acquisition unit **71** acquires the hypothetical

control content based on the generated target trajectory TR. The optimum operation content acquiring unit **71b** calculates the optimum operation content based on the hypothetical control content.

[0053] The driver operation content acquisition unit **73** acquires information on the driver operation content from the driver operation sensor **24**. The driver operation content includes an operation timing and an operation amount of the driver.

[0054] The notification determination unit **74** receives the optimum operation content from the control content acquisition unit **71** and the driver operation content from the driver operation content acquisition unit **73**. The notification determination unit **74** compares the driver operation content with the optimum operation content. The notification determination unit **74** determines whether to provide a notification including a feedback FB and the content of the feedback FB. As illustrated in FIG. **4**, the condition for notifying the feedback FB and the content of the feedback FB may be determined based on a comparison between the operation content of the drivers and a predetermined range including the optimum operation content. When the feedback FB is notified, the notification determination unit **74** performs notification via HMI unit **30**.

[0055] FIG. **6** is a block diagram illustrating a modification of the driver assistance system **10**. In the case of the example illustrated in FIG. **6**, the driver assistance system **10** always notifies the optimum operation content. Therefore, unlike the example of FIG. **5**, the driver operation content acquisition unit **73** and the notification determination unit **74** are unnecessary. In this case, the driver can always know the optimum operation content regardless of the operation content of the driver. The form of notification of the feedback FB is preferably such that the drivers do not feel troublesome. For example, it is preferable to notify by color change of the in-vehicle LED rather than by audio notification.

[0056] Further, the machine learning model may calculate the optimum operation content based on the information acquired by the sensor group **20**. By using the machine learning model in combination, it is possible to determine an optimum operation content in consideration of the driver's individual driving habits and the characteristics of the vehicle **1**.

[0057] In addition, at least some of the settings related to the feedback FB may be changeable by the drivers. For example, the first range R1 and the second range R2 may be adjustable according to the preference of the drivers. Through such a setting, the driver can appropriately adjust the frequency and content of the notification.

Claims

1. A driver assistance system for a driver of a vehicle, the driver assistance system comprising a control device that is equipped with a function of automated driving control, wherein the control device is configured to acquire a vehicle control content when assumption is made that the automated driving control will be performed, during manual driving of the vehicle by the driver, acquire a driver operation content that is a content of a driving operation of the vehicle by the driver, and notify the driver of feedback regarding the driving operation, based on a result of comparison between the vehicle control content and the driver operation content.
2. The driver assistance system according to claim 1, wherein: the driver operation content includes a driving operation timing; the vehicle control content includes a vehicle control timing; and when the driving operation timing is outside a first range including the vehicle control timing, the control device notifies the driver of the feedback indicating that the driving operation timing is not appropriate.
3. The driver assistance system according to claim 1, wherein: the driver operation content includes a driving operation amount; the vehicle control content includes a vehicle control amount; and when the driving operation amount is outside a second range including the vehicle control amount, the control device notifies the driver of the feedback indicating that the driving operation amount is

not appropriate.

4. The driver assistance system according to claim 1, wherein at least part of settings relating to the feedback is changeable by the driver.

5. A driver assistance system for a driver of a vehicle, the driver assistance system comprising a control device that is equipped with a function of automated driving control, wherein the control device is configured to acquire a vehicle control content when assumption is made that the automated driving control will be performed, during manual driving of the vehicle by the driver, and notify the driver of the vehicle control content.
